

Technical Documentation for ROBORACER GRAND PRIX

Team Falcons, Tongji University

Field	Description
Team	Falcons
Organization	Tongji University
Vehicle class	1:10 scale autonomous racing vehicle
Main computing unit	NVIDIA Jetson Orin Nano
Document purpose	Technical material submission for the vehicle hardware and software stack

1. Team and Vehicle Overview

Team Falcons from Tongji University is developing a 1:10 scale autonomous racing vehicle for the 29th ROBORACER Autonomous Racing Competition. The vehicle is built on a Traxxas Fiesta ST Rally VXL-3S platform and integrates onboard sensing, localization, planning, and control modules through a ROS 2-based software stack.

The autonomous driving system is designed to perform all computation onboard the vehicle. The NVIDIA Jetson Orin Nano serves as the single main computing unit and executes the perception, localization, planning, control, and logging processes. During normal autonomous driving, the vehicle does not depend on offboard computation or external data transmission for control decisions.

The current development focus is reliable autonomous lap completion, stable trajectory tracking, and safe obstacle avoidance. Pure Pursuit is used as the current baseline path-tracking controller. A Linear Quadratic Regulator controller is being evaluated as a candidate for improved tracking performance after further real-car testing.

2. Vehicle Hardware Stack

The vehicle uses a 1:10 scale electric racing platform with onboard compute, LiDAR, camera, inertial sensing, and electronic drive control.

Subsystem	Component
Chassis	Traxxas Fiesta ST Rally VXL-3S
Main computing unit	NVIDIA Jetson Orin Nano

LiDAR	ORADAR MS200k
Camera	ORBEC Astra Pro
IMU	WIT JY901B IMU
Motor	Traxxas Velineon 3500KV brushless motor
ESC	Traxxas VXL-3S ESC
Battery	3S LiPo x 1
Vehicle command format	Ackermann steering command interface

The ORADAR MS200k LiDAR is used as the primary range sensor for localization and obstacle detection. The ORBBEC Astra Pro is used only as a camera/depth sensing device; no onboard detection, recognition, or VIO output from the camera is used for autonomous driving. The WIT JY901B IMU provides inertial measurements for state estimation and motion diagnostics.

The powertrain uses a single Traxxas Velineon 3500KV brushless motor driven by a Traxxas VXL-3S ESC. The battery configuration is a single 3S LiPo pack, within the competition limit of 4S LiPo or lower. The vehicle will be equipped with the required detection box and foam bumper for race scenarios involving multiple vehicles.

3. Software Stack Architecture

The software system is based on ROS 2 and is organized as a modular autonomy stack. The major software layers are:

- Sensor and vehicle bringup layer: starts the LiDAR, camera, IMU, ESC interface, odometry, static transforms, and vehicle command conversion nodes.
- State estimation and localization layer: uses AMCL for map-based localization and EKF-based fusion for odometry and IMU information.
- Planning layer: processes the track map, generates a global reference trajectory, and provides a local obstacle-avoidance trajectory when obstacles are detected.
- Control layer: tracks the selected reference trajectory and publishes Ackermann steering and speed commands. Pure Pursuit is the current baseline controller, while LQR is maintained as an evaluation candidate.
- Command and actuation layer: converts controller outputs into Ackermann drive commands and sends them through the vehicle command pipeline to the ESC and steering servo.
- Logging and evaluation layer: records tracking logs and ROS topic data for offline evaluation, controller tuning, and diagnosis of localization, steering, and speed response.

Simplified flow: LiDAR / Camera / IMU / ESC odometry -> TF and sensor drivers -> EKF + AMCL localization -> global trajectory generation -> Frenet local planning when obstacles are present -> Pure Pursuit baseline controller or LQR candidate controller -> Ackermann command pipeline -> ESC and steering actuation.

4. Localization and State Estimation

The vehicle uses AMCL as the main localization algorithm. Given a known occupancy-grid map of the track, AMCL estimates the vehicle pose by matching LiDAR scans against the map. This pose is used by the planning and control modules to determine the vehicle position relative to the reference trajectory.

Wheel/ESC odometry and IMU measurements are fused through an EKF-based state estimation pipeline. The EKF improves short-term motion consistency and provides filtered odometry information for control and diagnostics. The system uses ROS TF frames to maintain consistent transforms between the map, odometry, base_link, LiDAR, camera, and IMU frames.

Before autonomous driving, the team verifies that localization is stable and that the estimated vehicle pose is consistent with the map and the visible track geometry.

5. Planning and Obstacle Avoidance

The global planning module generates a reference trajectory from the track map. This trajectory is used for normal lap driving when the track is free of obstacles. The planner output is represented as a sequence of poses or waypoints in the map or odometry frame and is consumed by the controller.

For obstacle avoidance, the team has implemented a Frenet-frame planner. The planner uses the global trajectory as a reference path and samples candidate local trajectories in Frenet coordinates. Candidate trajectories are checked against detected obstacles and track constraints. The selected trajectory is intended to be collision-free and dynamically trackable by the controller.

Obstacle information is currently derived from LiDAR data. The Frenet planner is designed to support avoidance of static obstacles and to provide a foundation for future handling of dynamic obstacles, including opponent vehicles. At the current stage, the Frenet avoidance module has been implemented but has not yet completed full validation on the physical vehicle.

6. Control Algorithms

The current baseline controller is Pure Pursuit. It tracks the reference trajectory by selecting a lookahead point ahead of the vehicle and computing the steering command required to drive toward that point. This controller is simple, robust, and suitable for early real-car validation.

The Pure Pursuit implementation supports conservative testing parameters, speed scheduling, and tracking logs. During initial vehicle testing, the target speed and lookahead distance are kept conservative to reduce risk while validating localization, trajectory quality, and actuation response.

The team is also evaluating an LQR controller as a candidate for improved trajectory tracking. The LQR controller models lateral and heading error dynamics and computes steering feedback based on weighted tracking errors and steering effort. It may provide better stability and smoother tracking at higher speed, but it has not yet been selected as the final controller.

7. Safety and Competition Compliance

The vehicle is designed with competition compliance and operational safety in mind.

- All autonomous computation is performed onboard the vehicle by the Jetson Orin Nano.

- During normal autonomous driving, no external control data is transmitted to the vehicle.
- The vehicle uses one main computing unit, one LiDAR sensor, one powertrain motor, and a 3S LiPo battery.
- The ORBBEC Astra Pro is not used for onboard detection, recognition, or VIO output.
- The vehicle will include the required detection box and foam bumper for multi-vehicle race scenarios.
- The toggle emergency brake is intended for inspection and permitted emergency or referee-directed situations, not for manual intervention during normal autonomous driving.
- Manual intervention is not part of normal autonomous driving and will only be used under permitted emergency or referee-directed conditions.
- The software stack is designed to operate independently of Wi-Fi quality during driving.

Rule-related item	Documented implementation
One main computing unit	NVIDIA Jetson Orin Nano
One LiDAR	ORADAR MS200k
Camera restrictions	ORBBEC Astra Pro used only for camera/depth sensing; no onboard detection or VIO output used
One powertrain motor	Traxxas Velineon 3500KV brushless motor
Battery limit	3S LiPo x 1, within 4S or lower limit
Other sensors	WIT JY901B IMU used for inertial measurement; no indoor GPS
Onboard computation	Localization, planning, control, and logging executed onboard
Normal-driving transmission	No external control data transmitted to the vehicle during normal driving

8. Current Development Status and Next Steps

The core vehicle hardware stack has been selected, and the ROS 2 autonomy software architecture is under active integration. The baseline localization, planning, and Pure Pursuit control pipeline is the primary path for initial real-car validation. The Frenet obstacle avoidance planner and LQR controller are implemented or under active development, but both require further real-car testing before being considered final competition-ready components.

The team will continue validating the system progressively, prioritizing safe operation, compliance with competition rules, and reliable autonomous driving before increasing speed or race aggressiveness. Final controller selection will be based on real-car robustness and tracking performance.